

EDITOR	: Ramesh Chopra
EDITORIAL CORRESPONDENCE	: Editorial Secretary Phone: 011-26810601; E-mail: editsec@efy.in (Technical queries: eflyab@efy.in)
SUBSCRIPTIONS & MISSING ISSUES	: Phone: 011-26810601 or 02 or 03 E-mail: support@efy.in
BACK ISSUES, BOOKS, CDs, PCBs etc.	: Kin's Spares, New Delhi Phone: 011-2637661, 2637662 E-mail: info@kintospares.com
NEWSSTAND DISTRIBUTION	: Ph: 011-40596600 E-mail: eflydc@efy.in
ADVERTISEMENTS NEW DELHI (HEAD OFFICE)	: Ph: 011-26810601 or 02 or 03 E-mail: eflymag@efy.in
MUMBAI	: Ph: 022-24950047, 24928520 E-mail: eflymum@efy.in
BENGALURU	: Ph: 080-25260394, 25260203 E-mail: eflybng@efy.in
PUNE	: Ph: 08800295610, 09870682995 E-mail: eflypune@efy.in
GUJARAT	: Ph: 079-61344948 E-mail: eflyahd@efy.in
CHINA	: Power Pioneer Group Inc. Ph: (86 755) 83729797, (86) 1392380295 E-mail: powerpioneer@efy.in
JAPAN	: Tandem Inc., Ph: 81-3-3541-4166 E-mail: tandem@efy.in
SINGAPORE	: Publicitas Singapore Pte Ltd Ph: +65-6836 2272 E-mail: publicitaz@efy.in
TAIWAN	: J.K. Media, Ph: 886-2-8726780 ext. 10 E-mail: jkmedia@efy.in
UNITED STATES	: E & Tech Media Ph: +1 860 536 6677 E-mail: veroniquelemaque@gmail.com

Printed, published and owned by Ramesh Chopra. Printed at International Press o-Pack Ltd, C-4 to C-11, Haryana Complex, Phase-II Extension, MODA-201 305, Gurgaon Road, New Delhi, India, on the first day of each month and published from P-4071, DDA Industrial Area, Phase-1, New Delhi 110053. Copyright 2016. All rights reserved through the publisher. No part of this magazine or any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any error due to publishing errors. Articles that cannot be used are returned to the author if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EPY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newsstand Price	You Pay	Digital	Overseas
Year	(₹)	(₹)	(₹)	Digital Print
Two	2400	1800	750	US\$ 15
One	1200	960	280	US\$ 5 US\$ 120

Please send payments only in favour of EPY Enterprises Pvt Ltd

Contents



TECH FOCUS

52 Is A Smart Home Safe Enough For You?

BUYERS' GUIDE

83 Oscilloscope: Get The Best At The Best Price



18 Information Technology

The Internet: Information From The Sky

22 Manufacture

High-Speed PCB Design For Electromagnetic Compatibility (Part 1 of 5)

28 Automotive

Vehicles' Ethernet Emerging Trends And Challenges

38 Design

Technologies That Drive Communication Devices

44 Information Technology

The Power Of Grid computing

62 Test & Measurement

From Spying To Dynamic Traffic Control With Thermal Cameras

66 Education

How Modern Technology Is Revolutionising The Academic Sector

76 Defence Electronics

Directed Energy Weapons: High-Power Microwaves (Part 4 of 8)

82 EFY Plus DVD

Blend Fine Artistry In Engineering And Design Through This DVD

90 Innovation

Car Head-Up Display Based Smart Infotainment System

92 Embedded

The Latest In Storage Products

116 Make In India

Market Survey: Will Home Automation Create A Bull Market?

eStyle

124 Buyers' Guide: Selecting The Best Wi-Fi Solution You Can Get Today

126 Do-It-Yourself: How To Clean Up Your Gmail Inbox

DO-IT-YOURSELF



- 94 Low-Cost Air-Quality Meter
- 97 SMS Based Home Automation Using Arduino
- 100 Lost Plane Finder
- 102 Audio Noise Limiter
- 104 Wireless Water-Level Indicator Without Microcontroller
- 107 Raspberry Pi Based SMS Server
- 110 Basic Image Processing Using MATLAB

EFY Plus DVD

CometCAD: From Symbol To PCB Layout I

PCB123 For Easier And Faster PCB Turnout IV

eCos For Limitless Configurations Of Your Embedded OS VII

Regulars

- 06 Feedback
- 08 Q&A
- 10 Useful Websites
- 12 Tech News
- 112 Make In India: Industry News
- 119 New Products
- 122 First Look
- 128 Business Pages Ads
- 141 Electronics Mart Ads
- 144 Product Categories Index + Attractions During 2016
- 145 Advertisers' Index

Interviews

87

NETWORKING:
"This Mesh Algorithm Allows OTA Upgrades" — Anand Bhandari, head - IoT/Telematics M2M, Maven Systems

88

POWER SUPPLY:
"Gel Batteries Are Much Smaller In Terms Of Size And Footprint" — Biju Bruno, managing director, Greenwin Technologies